

(possibly extended). Indeed, as  $ML$  and  $LZ$  are midlines in triangles  $QAB$ ,  $QBR$ , respectively, they are both parallel to  $AB$  and thus points  $M$ ,  $L$ , and  $Z$  are collinear. The collinearity of  $M$ ,  $X$ ,  $K$  and  $K$ ,  $Y$ ,  $L$  is proved analogously.

Now by Menelaus' Theorem in triangle  $KLM$  for points  $X$ ,  $Y$ , and  $Z$  we need to prove that

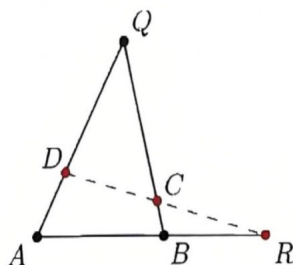
$$\frac{MX}{XK} \cdot \frac{KY}{YL} \cdot \frac{LZ}{ZM} = 1.$$

But recalling that the length of a midline is half the length of the corresponding side, these ratios may be “projected” on the sides of triangle  $ABQ$ . We have

$$\frac{MX}{XK} = \frac{QC}{CB}, \quad \frac{KY}{YL} = \frac{AD}{DQ}, \quad \frac{LZ}{ZM} = \frac{BR}{RA}.$$

This enables us to forget all the midpoints, since now it suffices to prove an identity concerning only points  $A$ ,  $B$ ,  $C$ ,  $D$ ,  $P$ , and  $Q$ . Namely

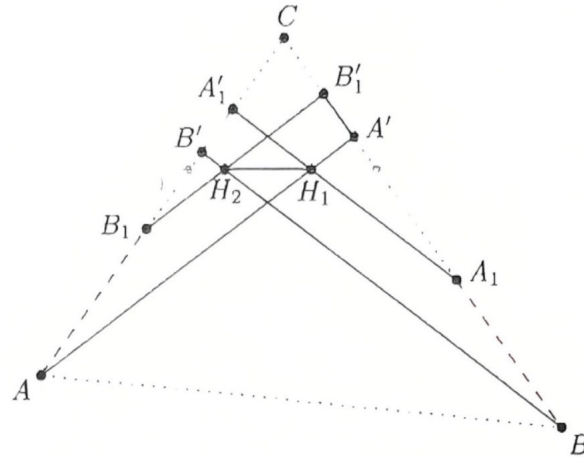
$$\frac{QC}{CB} \cdot \frac{BR}{RA} \cdot \frac{AD}{DQ} = 1.$$



We may celebrate, as the latter is true by Menelaus' Theorem applied to triangle  $ABQ$  and collinear points  $D$ ,  $C$ , and  $R$ .

39. [Moscow Math Olympiad 2009] In acute triangle  $ABC$  let  $A_1$ ,  $B_1$  be the points of tangency of  $A$ -excircle with  $BC$  and  $B$ -excircle with  $AC$ , respectively. Let  $H_1$ ,  $H_2$  be the orthocenters of triangles  $CAA_1$  and  $CBB_1$ , respectively. Prove that  $H_1H_2$  is perpendicular to the angle bisector of  $\angle ACB$ .

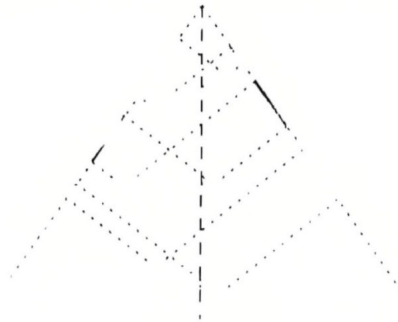
**Proof.** Let us view the orthocenter  $H_1$  as the intersection of altitudes  $AA'$  and  $A_1A'_1$  and the orthocenter  $H_2$  as the intersection of altitudes  $BB'$  and  $B_1B'_1$ . A reasonable way to make use of the fact that  $A_1$ ,  $B_1$  are the points of contact of the excircles is to recall  $AB_1 = \frac{1}{2}(a+b-c) = BA_1$  (see Proposition 1.15(c)). In fact, we will use only  $AB_1 = BA_1$ .



Note that the angle  $\angle C$  is contained by segments  $AB_1$  and  $A'B'_1$  and also by  $BA_1$  and  $B'A'_1$ . Thus, projecting the equal segments  $AB_1$ ,  $BA_1$  onto the lines  $BC$ ,  $AC$  we infer

$$A'B'_1 = AB_1 \cos \angle C = BA_1 \cos \angle C = B'A'_1.$$

Now focus on the segment  $H_1H_2$ . We have just shown that its projections to the lines  $AC$ ,  $BC$  are of the same length. The result should be apparent. Placing the bisector of  $\angle BCA$  vertically, it is quite plausible that if the segment  $H_1H_2$  was skew, one of its projections would be longer than the second one. Let us prove it rigorously.



Since  $\angle C$  is acute and the points  $C$ ,  $B_1$ ,  $A$  and  $C$ ,  $A_1$ ,  $B$  lie on the sides of  $\angle C$  in this order, line  $H_1H_2$  intersects the segments  $BC$ ,  $AC$ . Denote the intersections by  $X$ ,  $Y$ , respectively.

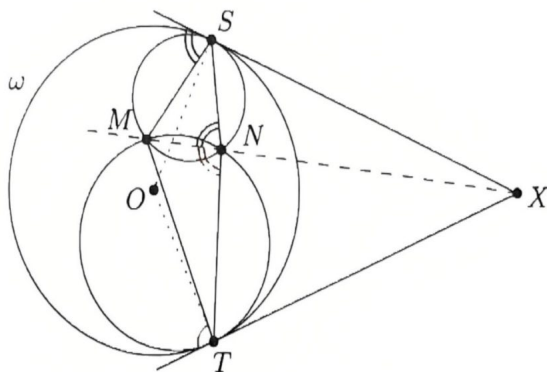
Similarly as before, we obtain

$$\cos \angle CXY = \frac{A'B'_1}{H_1H_2} = \frac{B'A'_1}{H_1H_2} = \cos \angle XYC.$$

Thus,  $\angle CXY = \angle XYC$  and the triangle  $CXY$  is isosceles. Its base is then indeed perpendicular to the angle bisector of  $\angle BCA$ .

40. [China 1997] A circle  $\omega$  with center  $O$  is internally tangent to two circles in its interior at points  $S$  and  $T$  which are not diametrically opposite. Suppose the two circles intersect at  $M$  and  $N$  with  $N$  closer to  $ST$ . Show that  $OM \perp MN$  if and only if  $S, N, T$  are collinear.

**Proof.** To make use of the tangency, draw common tangents at points  $S$  and  $T$  and denote by  $X$  their intersection. Since  $XS = XT$ , point  $X$  has equal power to the two small circles and thus it lies on their radical axis  $MN$ .

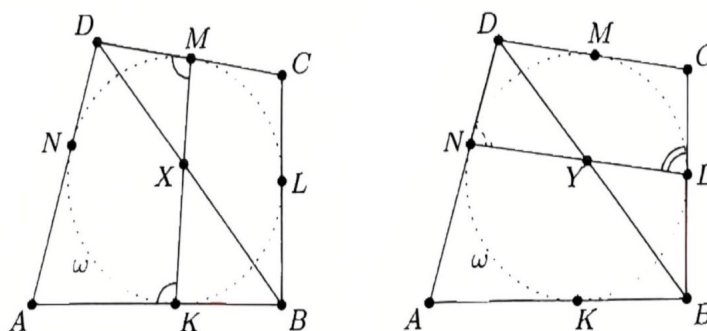


The condition  $OM \perp MN$  is equivalent to  $\angle OMX = 90^\circ$ . As points  $S$  and  $T$  both lie on a circle with diameter  $OX$ , the latter happens if and only if the points  $S, M, T$  and  $X$  are concyclic.

On the other hand, we have  $\angle SNM = 180^\circ - \angle MSX$  and  $\angle MNT = 180^\circ - \angle XTM$  (see Proposition 1.34). Summing these two relations we deduce that the points  $S, N, T$  are collinear if and only if  $SMTX$  is cyclic and thus we are done.

41. Let  $ABCD$  be a quadrilateral with an inscribed circle  $\omega$  and let the points of tangency of the incircle with sides  $AB, BC, CD, DA$  be  $K, L, M, N$ , respectively. Prove that the lines  $AC, BD, KM$ , and  $LN$  are concurrent.

**Proof.** We start by observing that  $\angle DMK = \angle MKA$ , as both angles correspond to the same arc  $MK$  of  $\omega$  (see Proposition 1.34). Our strategy will be ratios and the Law of Sines.



Let  $KM$  intersect  $BD$  at  $X$ . We intend to find the ratio in which  $X$  divides  $BD$ . The Law of Sines in triangles  $XMD$  and  $KBX$  yields

$$DX = MD \cdot \frac{\sin \angle DMX}{\sin \angle MXD}, \quad XB = KB \cdot \frac{\sin \angle KBX}{\sin \angle KXB}.$$

Since  $\sin \angle DMX = \sin \angle KBX$ , we can find the sought-after ratio as

$$\frac{DX}{XB} = \frac{MD}{KB}.$$

Next, we intersect  $BD$  and  $NL$  at point  $Y$  and find how it divides  $BD$ . After analogous calculation, we obtain

$$\frac{DY}{YB} = \frac{ND}{LB},$$

and we observe that due to Equal Tangents from  $B$  and  $D$ , the right-hand sides of the last two relations are equal and thus  $Y = X$  and  $BD$ ,  $KM$ , and  $LN$  are indeed concurrent.

In the same fashion, we show that also  $AC$ ,  $KM$ , and  $LN$  are concurrent.

#### 42. Orthologic triangles.

Let  $ABC$  and  $A'B'C'$  be two triangles in plane. Show that the perpendiculars from  $A'$  to  $BC$ , from  $B'$  to  $CA$  and from  $C'$  to  $AB$  (denote their feet by  $X$ ,  $Y$ , and  $Z$ , respectively) are concurrent if and only if the perpendiculars from  $A$  to  $B'C'$ , from  $B$  to  $C'A'$ , and from  $C$  to  $A'B'$  are concurrent.

**Proof.** By Carnot's Theorem (see Introductory Problem 49) the perpendiculars to the sides of triangle  $ABC$  are concurrent if and only if

$$BX^2 + CY^2 + AZ^2 = CX^2 + AY^2 + BZ^2.$$

Next, we rewrite this condition so that points  $A'$ ,  $B'$ , and  $C'$  are involved. From the perpendicularity criterion (see Proposition 1.22) for  $A'X \perp BC$  we learn that

$$BX^2 - CX^2 = A'B^2 - A'C^2.$$

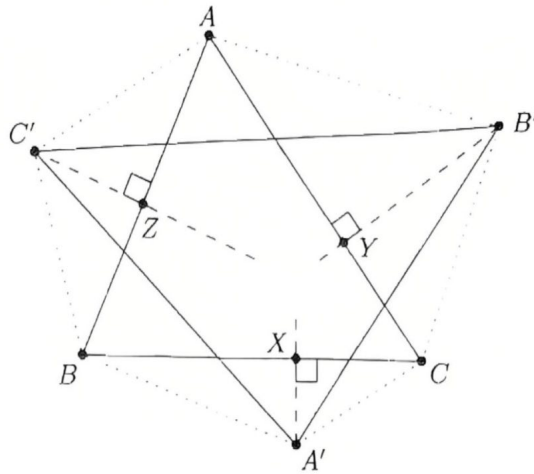
Similarly, we obtain

$$CY^2 - AY^2 = B'C^2 - B'A^2 \quad \text{and} \quad AZ^2 - BZ^2 = C'A^2 - C'B^2.$$

Thus the original condition can be equivalently rewritten as

$$A'B^2 + B'C^2 + C'A^2 = B'A^2 + C'B^2 + A'C^2.$$





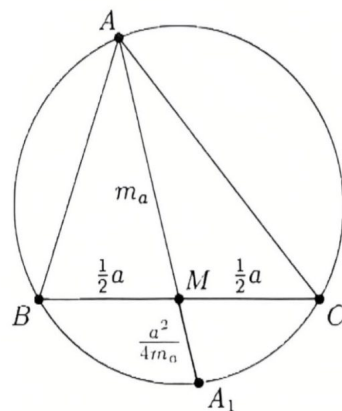
It remains to realize that the concurrence of the perpendiculars to the sides of triangle  $A'B'C'$  can be treated in the same way and turns out to be equivalent to the same condition, from which the conclusion follows.

43. [All-Russian Olympiad 1994] Let  $ABC$  be a triangle with medians  $m_a$ ,  $m_b$ ,  $m_c$  and circumradius  $R$ . Prove that

$$\frac{b^2 + c^2}{m_a} + \frac{c^2 + a^2}{m_b} + \frac{a^2 + b^2}{m_c} \leq 12R.$$

**Proof.** We avoid tough computations by geometric argument but it's going to be tricky! We divide the inequality by 2 and recall the median formula (see Corollary 1.24). Then we rewrite each term as follows

$$\frac{b^2 + c^2}{2m_a} = \frac{\frac{1}{2}(b^2 + c^2) - \frac{a^2}{4}}{m_a} + \frac{a^2}{4m_a} = m_a + \frac{a^2}{4m_a}.$$



The clever idea now is to apply Power of a Point. Indeed, denote by  $M$  the midpoint of  $BC$  and by  $A_1$  the point where the extended median

meets the circumcircle  $\omega$  for the second time. Then taking the power of  $M$  with respect to  $\omega$  gives

$$\frac{a^2}{4} = MB \cdot MC = m_a \cdot MA_1, \quad \text{hence} \quad MA_1 = \frac{a^2}{4m_a}.$$

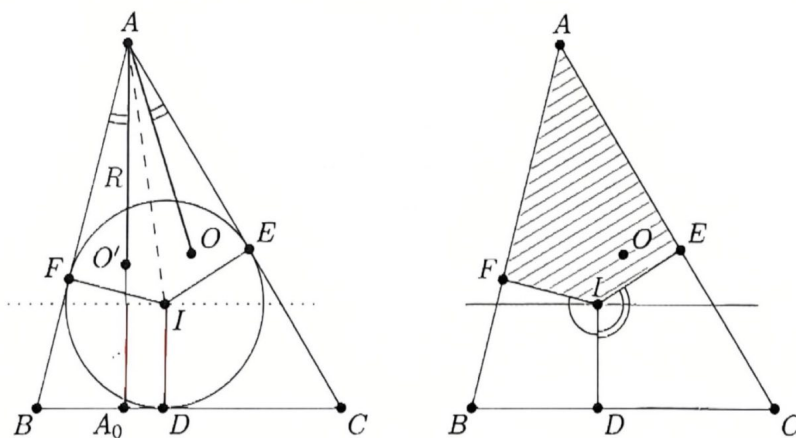
So in fact

$$\frac{b^2 + c^2}{2m_a} = m_a + MA_1 = AA_1 \leq 2R,$$

where the last inequality holds because diameter is the longest chord. Adding similar inequalities for the other two fractions gives the result.

44. [Paul Erdős] Show that in acute triangle  $ABC$  we have  $r + R \leq h$ , where  $r$ ,  $R$ , and  $h$  are the inradius, circumradius and the longest altitude, respectively.

**Proof.** Denote by  $I$  the incenter of triangle  $ABC$  and by  $D$ ,  $E$ ,  $F$  the points of contact of the incircle with the sides  $BC$ ,  $CA$ ,  $AB$ , respectively. Segments  $ID$ ,  $IE$ ,  $IF$  split triangle  $ABC$  into three quadrilaterals. Triangle  $ABC$  is acute, so its circumcenter  $O$  lies inside of



it. Without loss of generality suppose it lies inside  $AFIE$  (including the boundary), place  $BC$  horizontally with  $A$  on top and drop altitude  $AA_0$ .

It suffices to prove that  $r + R \leq AA_0$ . As  $O$  lies inside the quadrilateral  $AFIE$  which is symmetric about  $AI$ , so does its reflection  $O'$  about  $AI$ . Furthermore, since “ $H$  and  $O$  are friends” (see Proposition 1.47), point  $O'$  lies on the altitude  $AA_0$ .

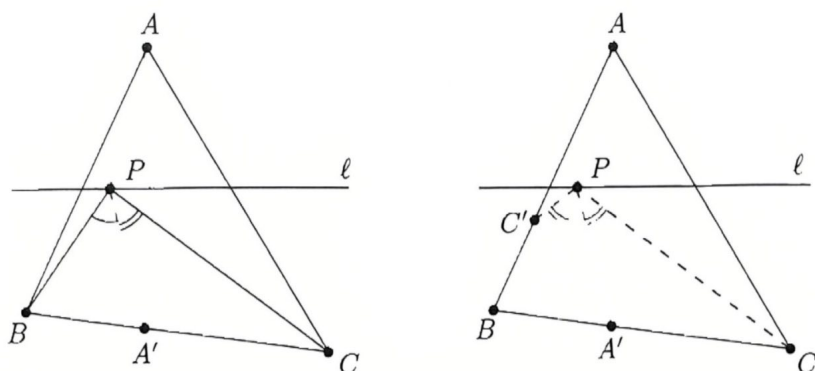
What remains to prove now is  $A_0O' \geq DI$ . But that’s obvious! Since both  $\angle B$  and  $\angle C$  are acute, both  $\angle DIE$  and  $\angle DIF$  are obtuse implying that point  $I$  is “the lowest” point of quadrilateral  $AFIE$ . Point  $O'$  (being inside  $AFIE$ ) is thus not below  $I$  and we may conclude.

45. [USAMO 2012, Titu Andreescu and Cosmin Pohoată] Let  $P$  be a point in the plane of triangle  $ABC$ , and  $\ell$  a line passing through  $P$ . Let  $A'$ ,  $B'$ ,  $C'$  be the points where the reflections of lines  $PA$ ,  $PB$ ,  $PC$  with respect to  $\ell$  intersect lines  $BC$ ,  $AC$ ,  $AB$  respectively. Prove that  $A'$ ,  $B'$ ,  $C'$  are collinear.

**Proof.** First we note that if any of the points  $A'$ ,  $B'$ ,  $C'$  coincides with one of the vertices, say  $A' = C$ , then lines  $AP$  and  $CP$  are symmetric with respect to  $\ell$  and thus also  $C' = A$ , which means  $A'$ ,  $B'$ ,  $C'$  all lie on  $AC$ .

For the general case we intend to use Menelaus' Theorem. We begin with the ratio  $BA'/A'C$  which we find from the Ratio Lemma (see Proposition 1.18) as

$$\frac{BA'}{A'C} = \frac{BP}{CP} \cdot \frac{\sin \angle BPA'}{\sin \angle A'PC}.$$



Now the crucial observation is that  $\angle(C'P, PA) = -\angle(CP, PA')$  since these angles correspond in symmetry with respect to  $\ell$ . Thus,

$$\sin \angle APC' = \sin \angle A'PC$$

and we may rewrite the Ratio Lemma as

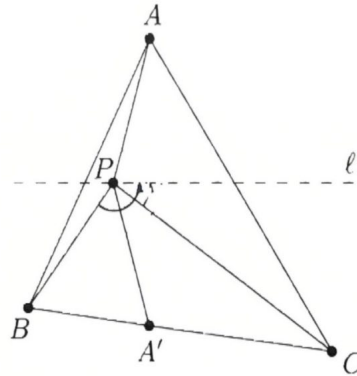
$$\frac{BA'}{A'C} = \frac{BP}{CP} \cdot \frac{\sin \angle BPA'}{\sin \angle APC'}$$

From here it is not hard to see that once we express the ratios  $CB'/B'A$  and  $AC'/C'B$  in the same manner and plug it in Menelaus' Theorem, everything cancels out. Indeed, we have

$$\begin{aligned} & \frac{BA'}{A'C} \cdot \frac{CB'}{B'A} \cdot \frac{AC'}{C'B} \\ &= \frac{BP}{CP} \cdot \frac{\sin \angle BPA'}{\sin \angle APC'} \cdot \frac{CP}{AP} \cdot \frac{\sin \angle CPB'}{\sin \angle BPA'} \cdot \frac{AP}{BP} \cdot \frac{\sin \angle APC'}{\sin \angle CPB'} \\ &= 1. \end{aligned}$$

One last thing we need to take care of is, since we used the undirected version of Menelaus' Theorem, to verify that either 0 or 2 points  $A'$ ,  $B'$ ,  $C'$  lie on the perimeter of triangle  $ABC$ . For this we use directed angles. If we denote by  $\alpha$ ,  $\beta$  and  $\gamma$  the angles  $\angle(AP, \ell)$ ,  $\angle(BP, \ell)$ ,  $\angle(CP, \ell)$ , respectively, we see that  $PA'$  lies between  $BP$  and  $CP$  (i.e.  $A'$  lies on the segment  $BC$ ) if and only if

$$\beta > 180^\circ - \alpha > \gamma, \quad \text{or equivalently} \quad \beta + \alpha > 180^\circ > \gamma + \alpha,$$



where the angles are no longer considered mod  $180^\circ$ . Analogous conditions hold for points  $B'$  and  $C'$  and it is easy to see that exactly 0 or 2 of them may be satisfied.

46. Let  $BC$  be the longest side of a scalene triangle  $ABC$ . Point  $K$  on the ray  $CA$  satisfies  $KC = BC$ . Similarly, point  $L$  on the ray  $BA$  satisfies  $BL = BC$ . Prove that  $KL$  is perpendicular to  $OI$  where  $O$ ,  $I$  denote the circumcenter and the incenter of triangle  $ABC$ , respectively.

**Proof.** Denote the circumcircle of triangle  $ABC$  by  $\Omega$  and its incircle by  $\omega$ . Since the locus of points  $X$ , for which the difference  $p(X, \Omega) - p(X, \omega)$  is equal to given constant, is a line perpendicular to  $OI$  (see Introductory Problem 53), it suffices to prove

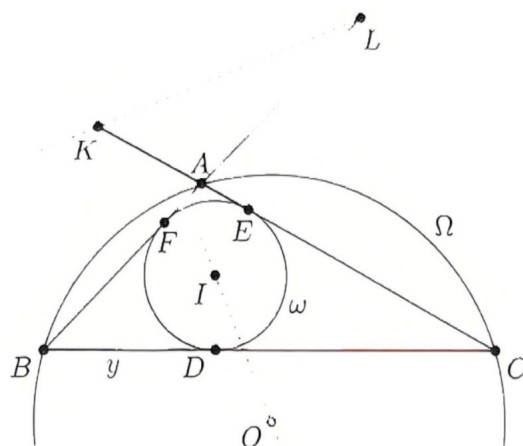
$$p(K, \Omega) - p(K, \omega) = p(L, \Omega) - p(L, \omega).$$

Denoting the points of contact of the incircle with the sides  $BC$ ,  $CA$ ,  $AB$  by  $D$ ,  $E$ ,  $F$ , respectively, all the powers can be expressed easily in the  $xyz$  notation, which reduces the whole problem to some straightforward algebra. We ease our lives a bit by noting  $KE = BD = y$  (symmetry about the angle bisector of  $\angle C$ ) and compute

$$\begin{aligned} p(K, \Omega) - p(K, \omega) &= KA \cdot KC - KE^2 = (y - x)(y + z) - y^2 \\ &= yz - x(y + z). \end{aligned}$$



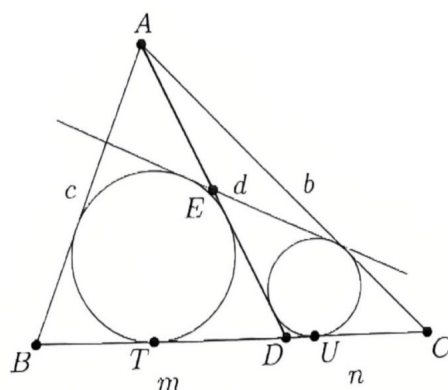
Since the last expression is symmetric with respect to  $y$  and  $z$ , the proof is complete.



47. [USAMO 1991] Let  $D$  be an arbitrary point on the side  $BC$  of a given triangle  $ABC$  and let  $E$  be the intersection of  $AD$  and the second external common tangent of the incircles of triangles  $ABD$  and  $ACD$ . As  $D$  assumes all positions between  $B$  and  $C$ , prove that the point  $E$  traces an arc of a circle.

**Proof.** First, when  $D$  is approaching  $C$ , it seems plausible that  $E$  is becoming the point of contact of the incircle of triangle  $ABC$  with the side  $AC$ . Likewise, when  $D$  tends to  $B$ , it appears that  $E$  tends to the point of contact of the incircle of triangle  $ABC$  with  $AB$ . Hence it is natural to expect that the desired locus is the arc of the circle with center  $A$  and radius  $x = \frac{1}{2}(b + c - a)$  (see Proposition 1.15) which lies inside the triangle  $ABC$ . Once we guessed it, it suffices to find  $AE$  in terms of the side lengths  $a, b, c$  only (it should turn out to be independent on the choice of  $D$ ).

Denote the points of contact of the incircles of triangles  $ABD, ACD$  with the side  $BC$  by  $T, U$ , respectively, the distances from  $D$  to the vertices by  $DA = d, DB = m, DC = n$ , and finally recall that  $DE = TU$  (see Proposition 1.13 (b)).



Hence

$$AE = d - ED = d - TU = d - (DT + DU).$$

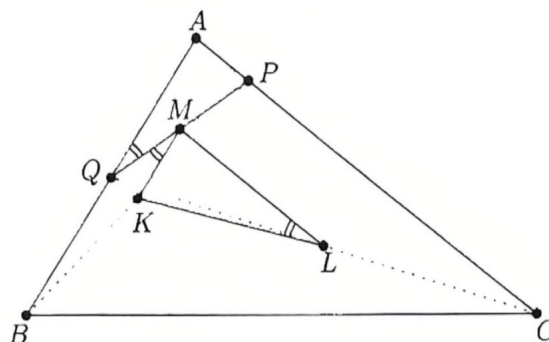
As  $DT = \frac{1}{2}(d + m - c)$  and  $DU = \frac{1}{2}(d + n - b)$ , we have

$$AE = \frac{1}{2}(b + c - m - n) = \frac{1}{2}(b + c - a)$$

as desired.

48. [IMO 2009] Let  $ABC$  be a triangle with circumcenter  $O$ . The points  $P$  and  $Q$  are interior points of the sides  $CA$  and  $AB$ , respectively. Let  $K$ ,  $L$  and  $M$  be the midpoints of the segments  $BP$ ,  $CQ$  and  $PQ$ , respectively, and let  $\Gamma$  be the circle passing through  $K$ ,  $L$ , and  $M$ . Suppose that the line  $PQ$  is tangent to the circle  $\Gamma$ . Prove that  $OP = OQ$ .

**Proof.** Without even drawing the circle  $\Gamma$ , we translate the tangency as  $\angle MLK = \angle QMK$  and  $\angle LKM = \angle LMP$  (see Proposition 1.34).



Next, we note that  $MK$  is the midline in triangle  $BPQ$  and  $ML$  is the midline in triangle  $CQP$ . Hence we can chase angles a little more

$$\angle MLK = \angle QMK = \angle MQA,$$

and similarly we obtain  $\angle LKM = \angle APQ$  and thus triangles  $AQP$  and  $MLK$  are similar (AA). From the ratios we learn that

$$\frac{AP}{AQ} = \frac{MK}{ML} = \frac{\frac{1}{2}QB}{\frac{1}{2}PC},$$

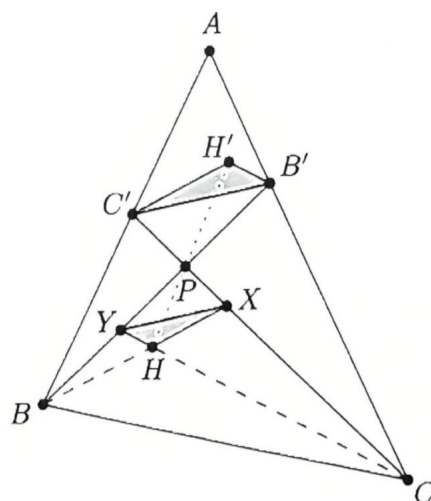
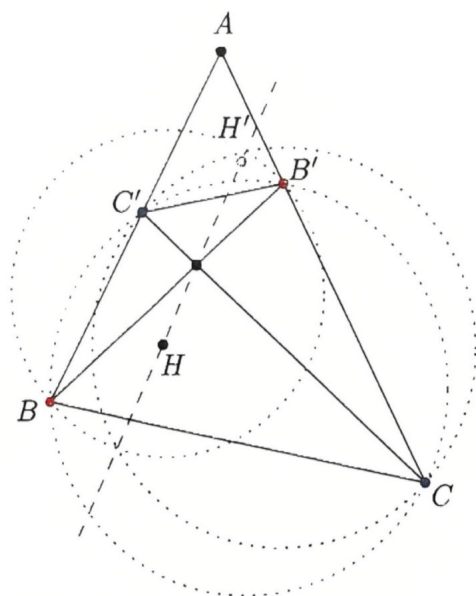
which rewrites as  $AP \cdot PC = AQ \cdot QB$ . But this implies that  $P$  and  $Q$  have the same power with respect to the circumcircle of triangle  $ABC$ , therefore they have the same distance from its center  $O$  (see Proposition 1.40(a)). We are done.

49. [IMO 1995 shortlist] Let  $ABC$  be a non-right triangle. A circle  $\omega$  passing through  $B$  and  $C$  intersects the sides  $AB$  and  $AC$  again at  $C'$  and  $B'$ ,

respectively. Prove that  $BB'$ ,  $CC'$  and  $HH'$  are concurrent, where  $H$  and  $H'$  are the orthocenters of triangles  $ABC$  and  $AB'C'$ , respectively.

**First Proof.** We recall Introductory Problem 51 and apply it for triangle  $ABC$  with orthocenter  $H$  and cevians  $BB'$  and  $CC'$  and also for triangle  $AB'C'$  with orthocenter  $H'$  and cevians  $B'B$  and  $C'C$ . We learn that the line  $HH'$  is in fact the radical axis of circles with diameters  $BB'$  and  $CC'$  (call them  $\omega_b$  and  $\omega_c$ ).

Moreover, we observe that  $BB'$  is precisely the radical axis of  $\omega$  and  $\omega_b$ , and  $CC'$  is the radical axis of  $\omega$  and  $\omega_c$ . Since pairwise radical axes of three circles intersect at their radical center (see Proposition 1.42), we are done.



**Second Proof.** Let  $P = BB' \cap CC'$  and also let  $X = BH \cap CC'$  and  $Y = CH \cap BB'$ . Antiparallelism is the key ingredient in this solution.

Perpendiculars  $BH$  and  $CH$  to  $AC$  and  $AB$  are antiparallel in  $\angle BAC$  and since  $BCB'C'$  is cyclic, they are also antiparallel in  $\angle BPC$  (see Corollary 1.36). Thus, in  $\angle BPC$  both  $B'C'$  and  $XY$  are antiparallel to  $BC$  implying that  $XY \parallel B'C'$ .

Furthermore, since  $BH \parallel C'H'$  (perpendiculars to  $AC$ ) and  $CH \parallel B'H'$  (perpendiculars to  $AB$ ) the triangles  $HXY$  and  $H'C'B'$  have corresponding sides parallel and hence the quadrilaterals  $PYHX$  and  $PB'H'C'$  are similar. Since the segments  $PH$  and  $PH'$  correspond, they are also parallel, thus  $HH'$  passes through  $P$  and we may conclude.

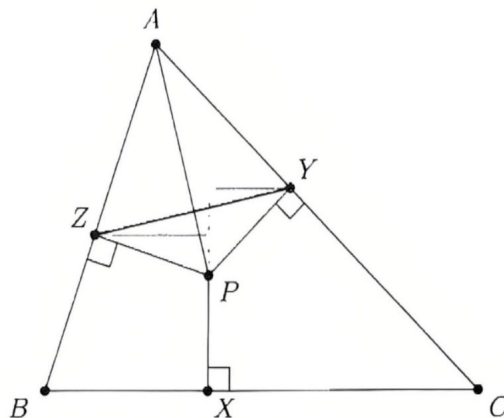
50. [USA TST 2000, Titu Andreescu] Let  $P$  be a point in the interior of triangle  $ABC$  with circumradius  $R$ . Prove that

$$\frac{AP}{a^2} + \frac{BP}{b^2} + \frac{CP}{c^2} \geq \frac{1}{R}.$$

**Proof.** Denote the projections of  $P$  onto  $BC$ ,  $CA$ ,  $AB$  by  $X$ ,  $Y$ ,  $Z$ , respectively, and recall the “key ingredient of the proof of the Erdős-Mordell inequality”, namely the inequality

$$PA \sin \angle A \geq PY \sin \angle C + PZ \sin \angle B,$$

which compares the length of  $YZ$  to the length of its projection onto  $BC$ .



Expressing every sine from the Extended Law of Sines in triangle  $ABC$  and cancelling  $2R$  this rewrites as

$$aPA \geq cPY + bPZ$$

or

$$\frac{PA}{a^2} \geq PY \cdot \frac{c}{a^3} + PZ \cdot \frac{b}{a^3}.$$

Likewise, we obtain

$$\begin{aligned} \frac{PB}{b^2} &\geq PZ \cdot \frac{a}{b^3} + PX \cdot \frac{c}{b^3}, \\ \frac{PC}{c^2} &\geq PX \cdot \frac{b}{c^3} + PY \cdot \frac{a}{c^3}. \end{aligned}$$

Summing these inequalities, applying AM-GM inequality to the terms in parentheses, and finally recalling the area formula involving circumradius (see Proposition 1.25) we can estimate the left-hand side (*LHS*) of the given inequality as

$$\begin{aligned} LHS &\geq PX \left( \frac{b}{c^3} + \frac{c}{b^3} \right) + PY \left( \frac{c}{a^3} + \frac{a}{c^3} \right) + PZ \left( \frac{a}{b^3} + \frac{b}{a^3} \right) \\ &\geq \frac{2 \cdot PX}{bc} + \frac{2 \cdot PY}{ca} + \frac{2 \cdot PZ}{ab} = \frac{4[ABC]}{abc} = \frac{1}{R}, \end{aligned}$$

which is exactly what we wanted.

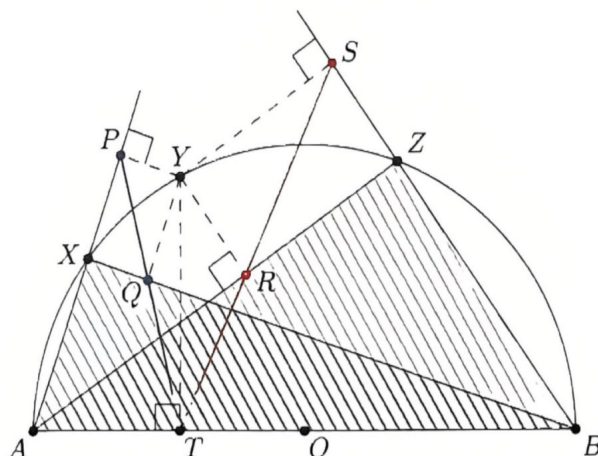


Equality in the first step requires  $YZ$  to be parallel to  $BC$  and so on. This occurs if and only if  $P$  is the circumcenter of triangle  $ABC$ . Equality in AM-GM requires  $a = b = c$ . Thus the equality holds if and only if triangle  $ABC$  is equilateral and  $P$  is its center.

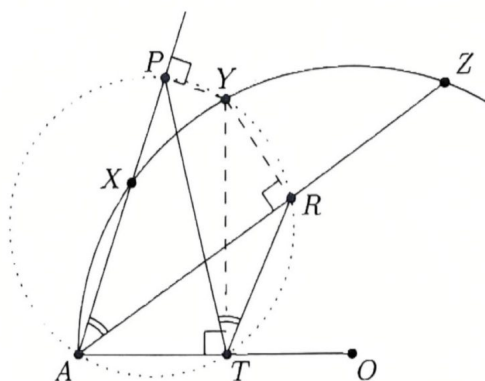
51. [USAMO 2010, Titu Andreescu] Let  $AXYZB$  be a convex pentagon inscribed in a semicircle of diameter  $AB$ . Denote by  $P, Q, R, S$  the feet of the perpendiculars from  $Y$  onto lines  $AX, BX, AZ, BZ$ , respectively. Prove that the acute angle formed by lines  $PQ$  and  $RS$  is half the size of  $\angle ZOX$ , where  $O$  is the midpoint of the segment  $AB$ .

**Proof.** A line passing through two feet of perpendiculars should remind us of the Simson line (see Example 1.15).

Denote by  $T$  the foot of perpendicular from  $Y$  onto line  $AB$ . Line  $PQ$  is the Simson line of point  $Y$  with respect to triangle  $ABX$ , hence it passes through  $T$ . Analogous reasoning shows that  $T \in RS$ .



On the other hand,  $\angle ZOX$  is the central angle corresponding to the minor arc  $XZ$ . Its half is therefore simply the corresponding inscribed angle. Thus, it is enough to show that  $\angle RTP = \angle RAP$  which is easily accomplished as points  $A, T, R, P$  all lie on a circle with diameter  $AY$ .



52. [Japan 2012] Let  $PAB$  and  $PCD$  be triangles such that  $PA = PB$ ,  $PC = PD$ , and triads of points  $P, A, C$  and  $B, P, D$  are both collinear in this order. A circle  $\omega_1$  passing through  $A$  and  $C$  intersects a circle  $\omega_2$  passing through  $B$  and  $D$  at distinct points  $X, Y$ . Prove that the circumcenter of the triangle  $PXY$  is the midpoint of the segment formed by the centers  $O_1, O_2$  of  $\omega_1, \omega_2$ , respectively.

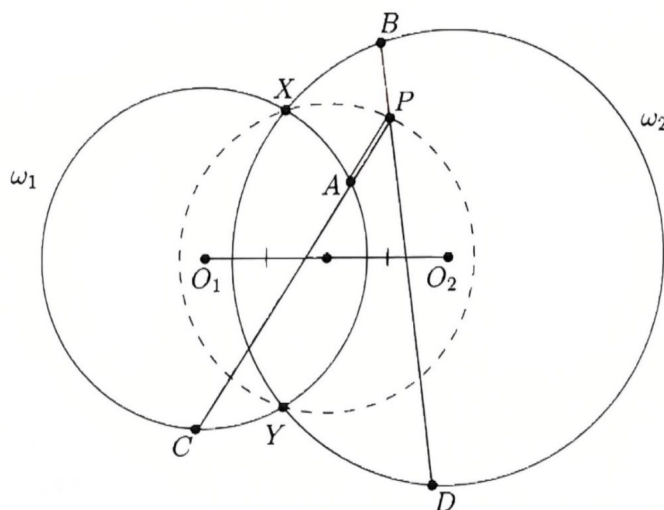
**Proof.** Believe it or not, this is going to be a one-sentence proof:

Recalling that the locus of points with fixed sum of powers with respect to two given circles is a circle centered at the midpoint of the centers of the two circles (see Introductory Problem 53 (b)), it suffices to observe that

$$p(X, \omega_1) + p(X, \omega_2) = 0 + 0 = 0, \quad p(Y, \omega_1) + p(Y, \omega_2) = 0 + 0 = 0,$$

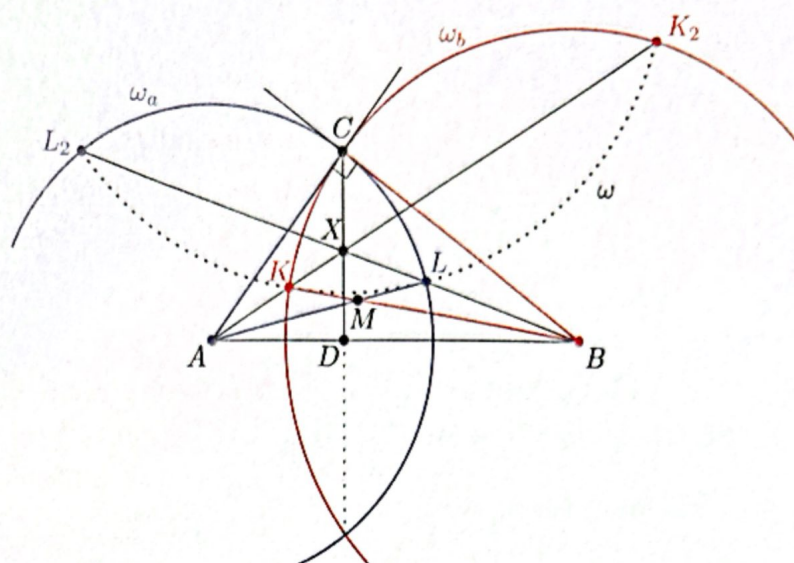
and

$$p(P, \omega_1) + p(P, \omega_2) = PA \cdot PC + (-PB \cdot PD) = 0.$$



53. [IMO 2012, Josef Tkadlec] Let  $ABC$  be a triangle with  $\angle BCA = 90^\circ$ , and let  $D$  be the foot of the altitude from  $C$ . Let  $X$  be a point in the interior of the segment  $CD$ . Let  $K$  be the point on the segment  $AX$  such that  $BK = BC$ . Similarly, let  $L$  be the point on the segment  $BX$  such that  $AL = AC$ . Let  $M$  be the point of intersection of  $AL$  and  $BK$ . Show that  $MK = ML$ .

**Proof.** Let  $\omega_a$  and  $\omega_b$  be the circles with centers  $A$  and  $B$ , respectively, passing through  $L$  and  $K$ , respectively. Since  $BC = BK$  and  $AC = AL$ , both these circles pass through  $C$  and as the angle by  $C$  is right,  $\omega_a$  is tangent to  $BC$  and  $\omega_b$  to  $AC$ . Moreover, the radical axis of  $\omega_a$  and  $\omega_b$  is the line passing through  $C$  perpendicular to  $AB$ , i.e. the altitude  $CD$ .



Hence if we let  $AX$  meet  $\omega_b$  for the second time at  $K_2$  and likewise  $BX$  meet  $\omega_a$  again at  $L_2$  then the Radical Lemma (see Proposition 1.43) implies that the quadrilateral  $L_2KLK_2$  is cyclic. Denote its circumcircle by  $\omega$ . Now the power of  $A$  with respect to  $\omega_b$  gives

$$AK \cdot AK_2 = AC^2 = AL^2,$$

thus  $AL$  is tangent to  $\omega$  (and so is  $ML$ ). Analogously,  $MK$  is tangent to  $\omega$ . Hence  $MK = ML$  by Equal Tangents.